Sustainable Energy Group Design Project

Group 16

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About our project

Gear
Turbine

AC Generator
AC-DC Conversion Circuit
Arduino

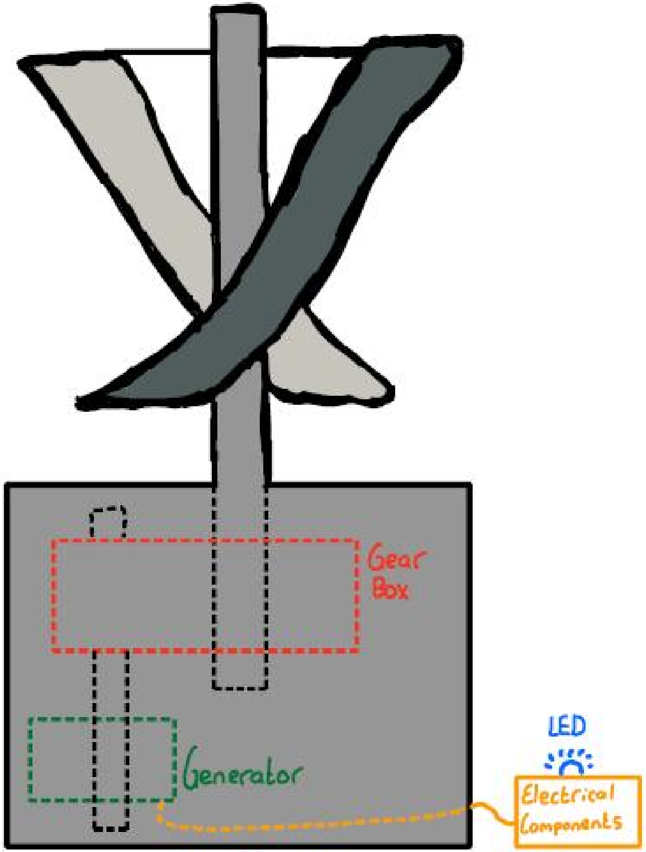
01

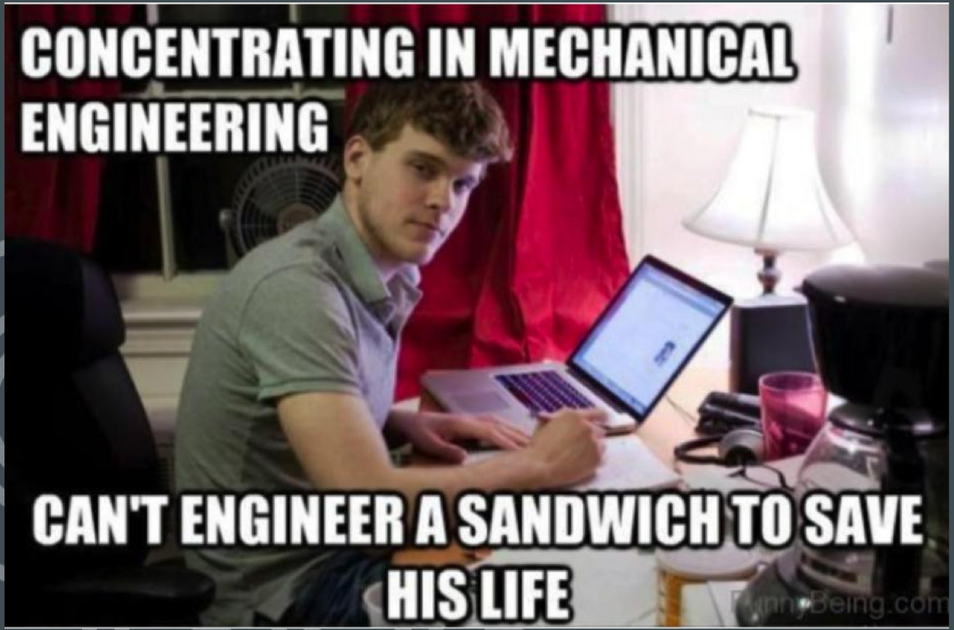
Introduction

Aim of Project

02

Overall Design



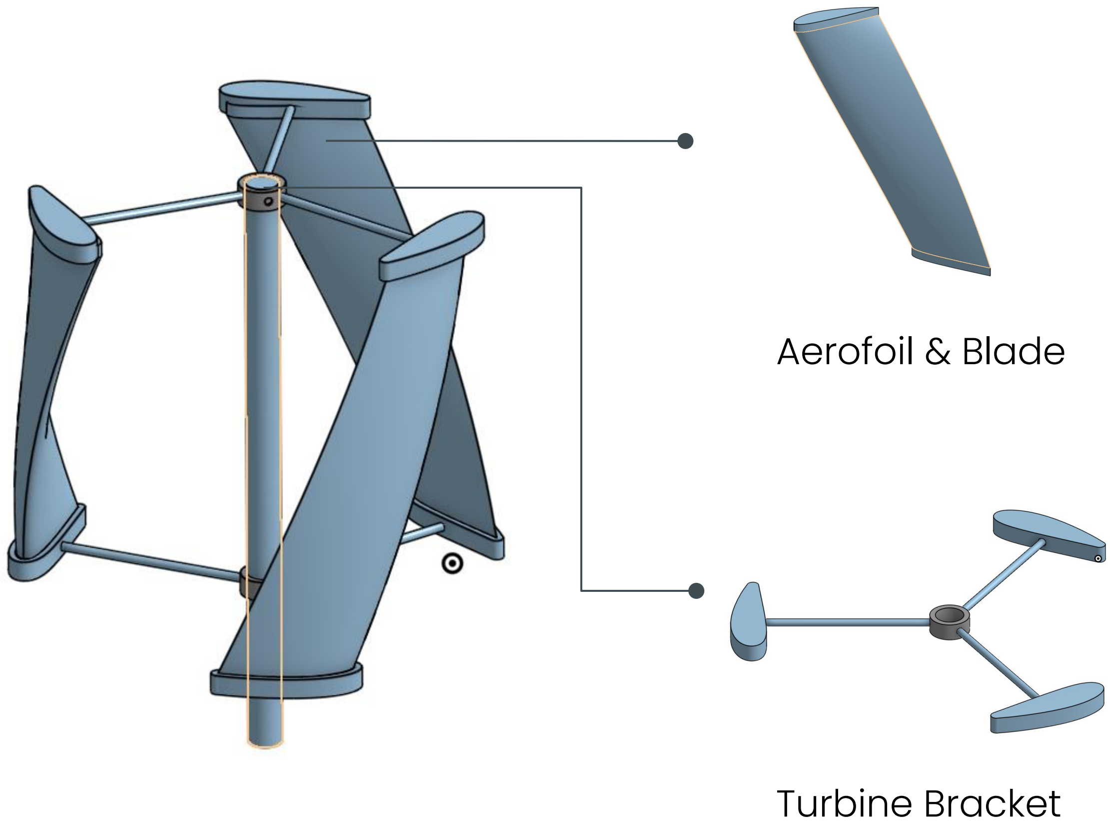


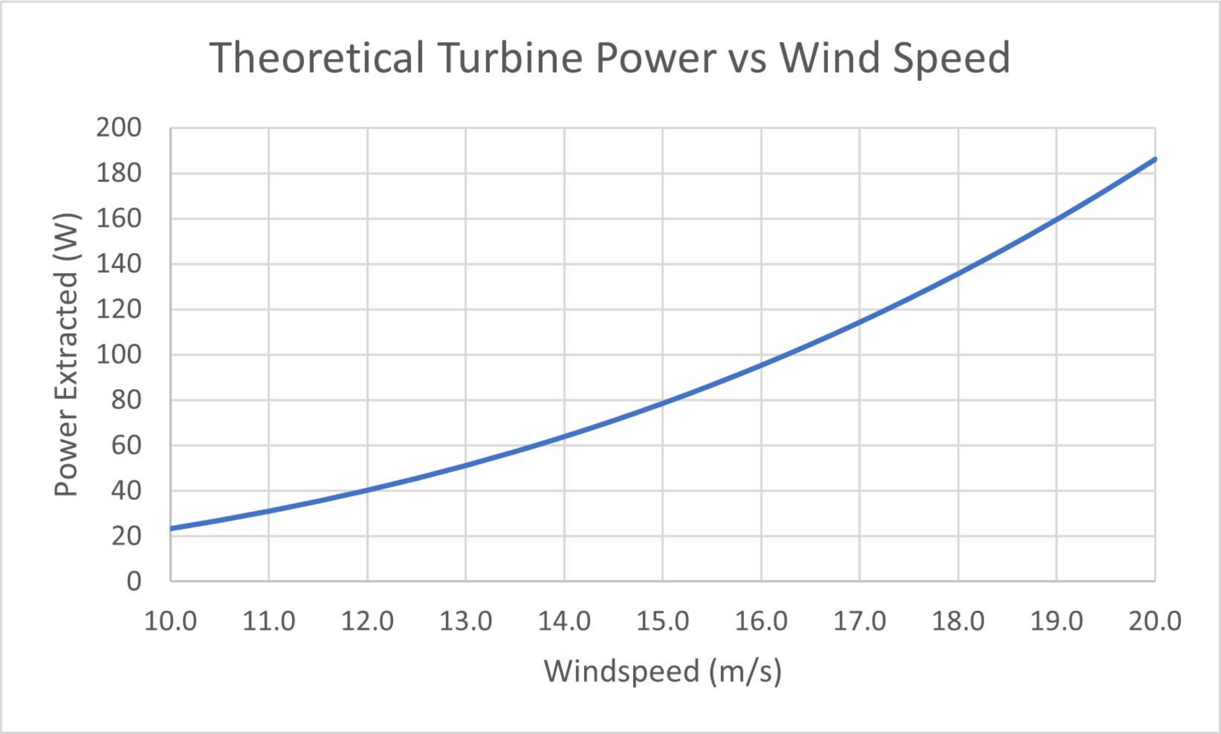
03

Mechanical Design

Turbine

- Calculation
- Aerofoil
- Blade
- Bracket





Theoretical Turbine Power vs Wind Speed
Power Extracted (W)
200
180
160
140
120
100
80
60
40
20
0
10.0
11.0
12.0
13.0
14.0
15.0
16.0
17.0
18.0
19.0
20.0
Windspeed (m/s)

- NACA 4421
- 45 degree helical blade
- Tip Speed Ratio: 1.5
- Turbine power coefficient: 0.44
- Diffuser not considered

20 m/s

Simulation Type

Aerodynamic Simulation

Time

Time : 0.159 of 0.196 s

Timestep : 809 of 998

Storing data from: 0.000 s

Performance

Power : 0.19 kW

Torque : 0.00 kNm

Rotor RPM : 1909.9 rpm

CP : 0.44

CT : 1.50

Tip Speed Ratio : 1.5

Wind Conditions

Wind Speed at Hub : 20.00 m/s

Uniform Wind

10 m/s

Simulation Type

Aerodynamic Simulation

Time

Time : 0.145 of 0.196 s

Timestep : 737 of 998

Storing data from: 0.000 s

Performance

Power : 0.02 kW

Torque : 0.00 kNm

Rotor RPM : 954.9 rpm

CP : 0.44

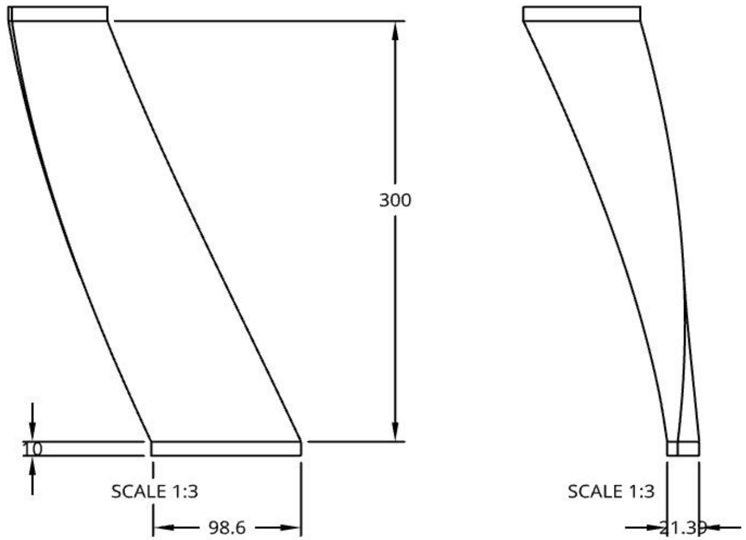
CT : 1.50

Tip Speed Ratio : 1.5

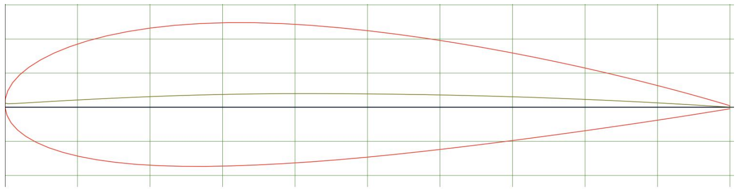
Wind Conditions

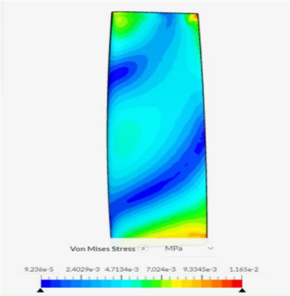
Wind Speed at Hub : 10.00 m/s

Uniform Wind

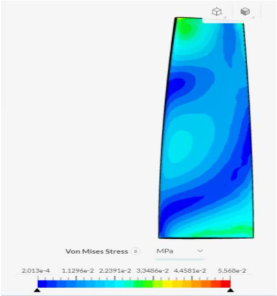


- Blade length: 300mm
- Chord: 100mm
- Aerofoil: NACA 4421

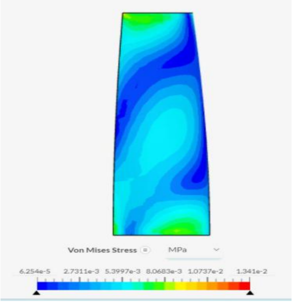




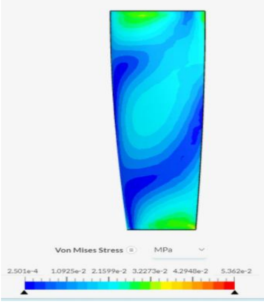
Back of blade 10m/s



Back of blade 20m/s

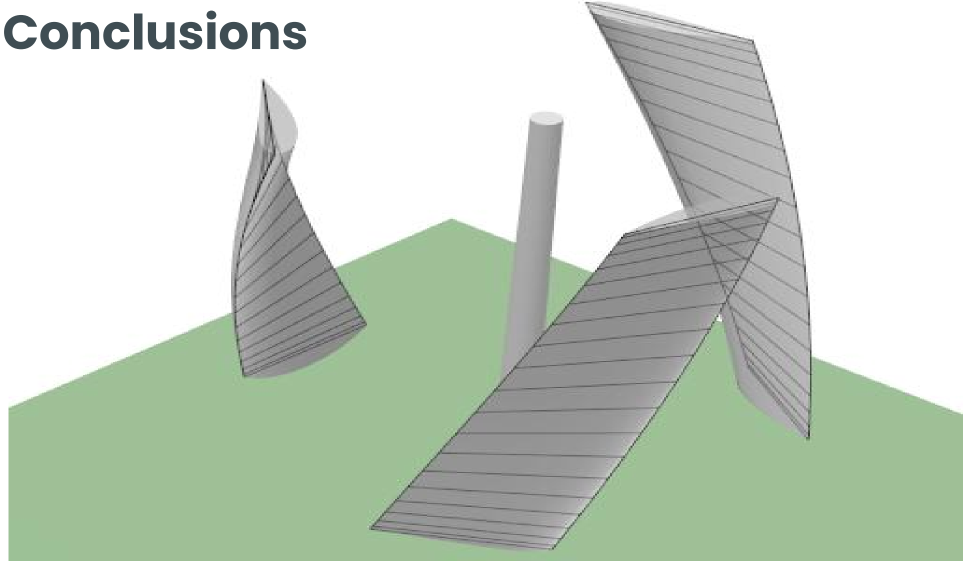


Front of blade 10m/s



Front of blade 20m/s

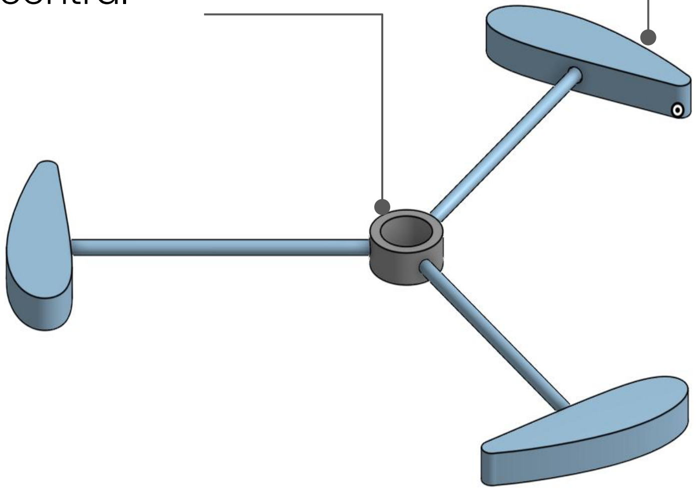
Conclusions

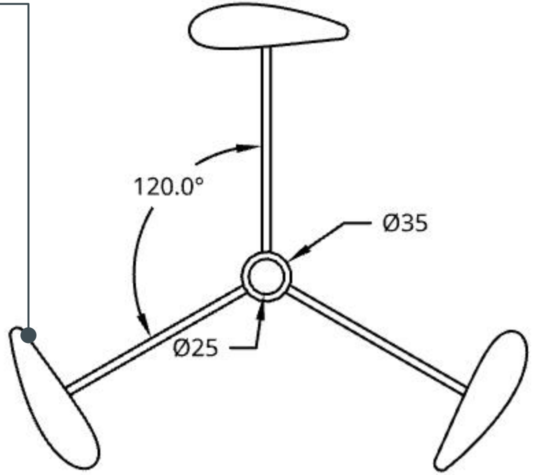


20m/s, largest stress: 0.056MPA < 32MPA
F: 2N at 10m/s, 7.6N at 20m/s
High tensile strength duct tape

Caps to be pressed on top the straight extrusion on the end of the blade

Collar that fits onto central shaft





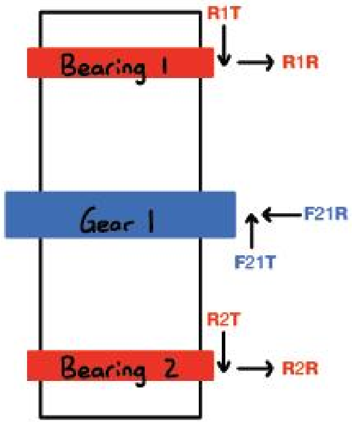
Top

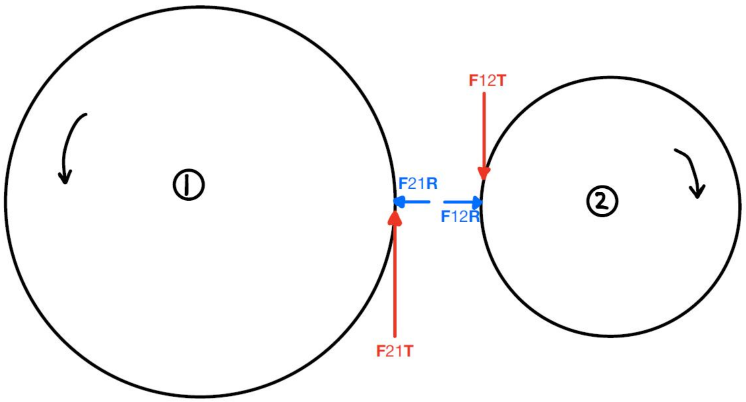
- Gear relationships
- Gearbox

Gear	Teeth	Pitch Diameter/ mm	Bore Diameter/ mm
1	70	70	25
2	24	24	10

Gear Ratio was found using the relationship: $GR = \frac{product\ of\ driven}{product\ of\ driver} = \frac{70}{24} \Longrightarrow GR_{ideal} = 2.92$

$$GR_{real} = \eta * GR_{ideal} \Longrightarrow GR_{real} = 2.33\ .$$





$$\Sigma Fy = 0 \implies Wrod = R2y \implies R2y = 10.6\ N$$

$$\Sigma F^R = 0 \implies R1^R + R2^R = F_{21}^R = 5.8\ N\ (1)$$

$$\Sigma F^T = 0 \implies R1^T + R2^T = F_{21}^T = 16\ N\ (2)$$

$$\Sigma M^R = 0 \implies R1^R * 0.017 + R2^R * 0.074 = F_{21}^R * 0.041 = 0.2378\ Nm\ \ (3)$$

$$\Sigma M^T = 0 \implies R1^T * 0.017 + R2^T * 0.074 = F_{21}^T * 0.041 = 0.2378\ Nm\ (4)$$

$$F_{21}^t = \frac{T_{input}}{r_{G1}} \implies F_{21}^t = 16\ N$$

Then, since pressure angle is assumed 20°, $F_{21}^r = F_{21}^t \cdot \tan(20) = 5.8\ N$ where F_{21}^r represents the force from Gear 2 onto Gear 1 in the radial direction and F_{21}^t represents the force from Gear 2 onto Gear 1 in the tangential direction.

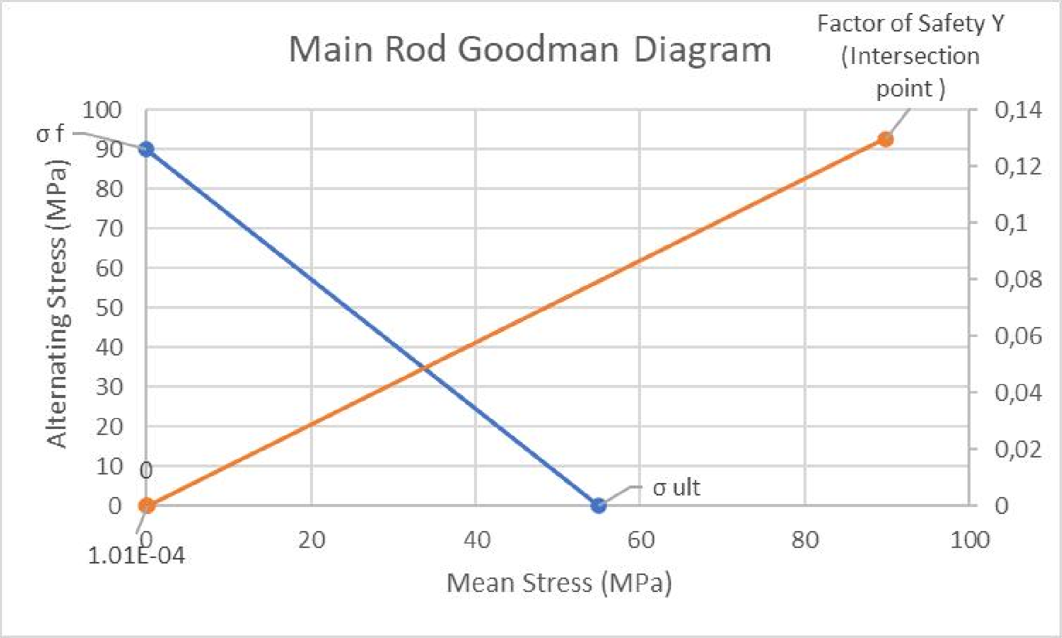
Due to action-reaction principle, $F_{12}^t = -F_{21}^t = -16\ N$ and $F_{12}^r = -F_{21}^r = -5.8\ N$

By solving as a simultaneous system, the equations (1), (2), (3) and (4), we get:

$$R1^R = 3.35\ N, \qquad R2^R = 2.44\ N\ , \qquad R1^T = 9.26\ N\ and\ \ R2^T = 6.74\ N$$

Applying similar method, we get for the second shaft:

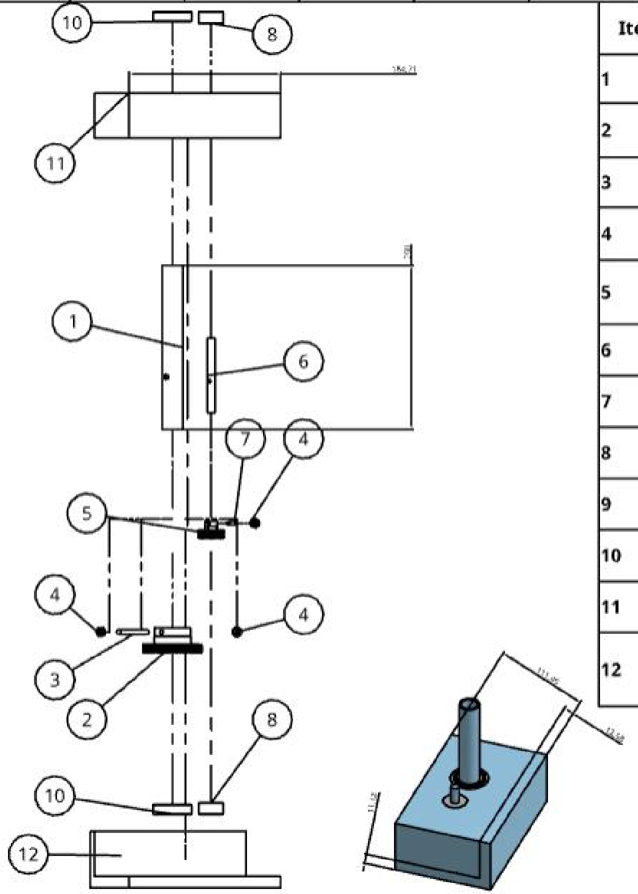
$$R3^R = -\ 3.13\ N, \qquad R4^R = -2.76\ N\ , \qquad R3^T = -8.6\ N\ and\ \ R4^T = -7.37\ N$$



$$\sigma_{em} = \frac{T * r}{J} = \frac{0.2\,(Nm) * 12.5 * 10^{-3}(m)}{\frac{\pi * (25 * 10^{-3})^4 (m)}{32}} \Rightarrow \sigma_{em} = 0.07\,MPa$$

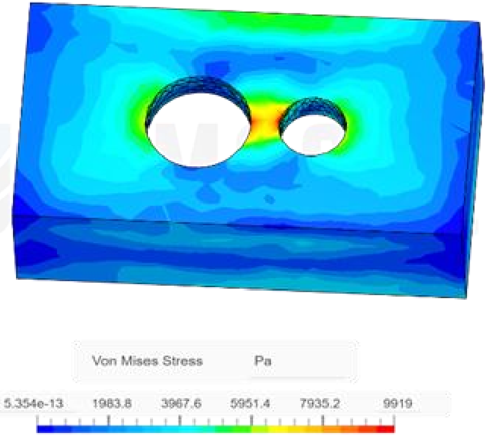
$$\sigma_{ea} = \frac{My}{I} = \frac{9.87\,(N) * 0.017\,(m) * 12.5 * 10^{-3}(m)}{\frac{1}{2} * 0.266\,(kg) * (12.5 * 10^{-3})^2 (m)} \Rightarrow \sigma_{ea} = 1.01 * 10^{-4}\,MPa$$

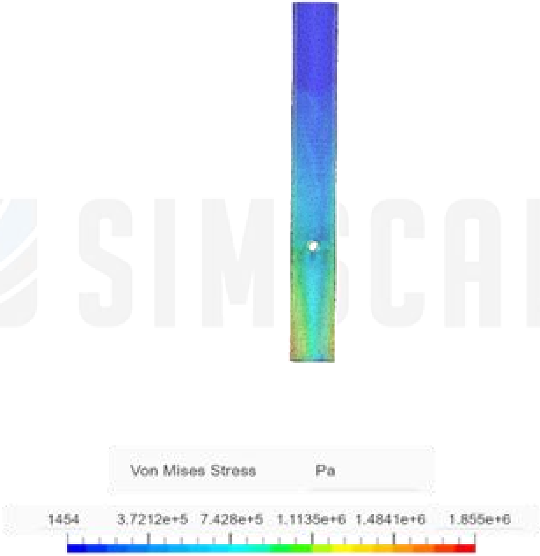
Hence, From graph: $Fatigue\ Factor\ Y = 0.13\ MPa$

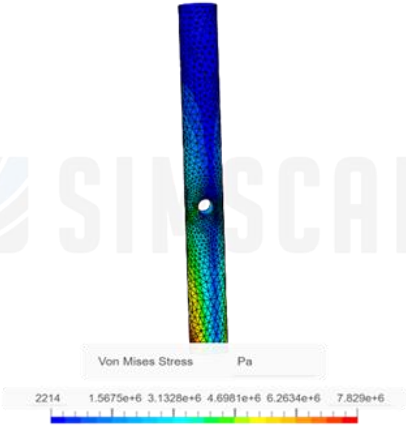


Item	Name	Quantity	Description
1	Main Rod (25mm)	1	Cost
2	MSGB1-70 adjusted (gear 1)	1	None
3	M6 Threaded Rod	1	0.10 £
4	M6 Nyloc Nut	3	0.15£
5	MSGA1-24 (gear 2)	1	3D Printed
6	10 mm metal bar	1	0.06£
7	M4 Threaded Rod	1	0.05£
8	Ballbearing 10	2	2.55£
9	25 x 47 Ball Bearing	2	2.55£
10	25 x 47 Ball Bearing	2	-
11	Lid part	1	0.80£
12	Main Case	1	2.00£

- Total cost:8.26 pounds


Von Mises Stress Pa
5.354e-13 1983.8 3967.6 5951.4 7935.2 9919


Von Mises Stress Pa
1454 3.7212e+5 7.428e+5 1.1135e+6 1.4841e+6 1.855e+6

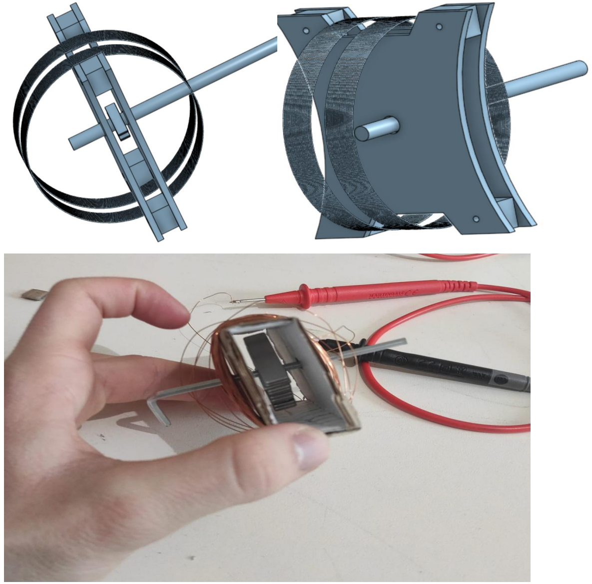

Von Mises Stress Pa
2214 1.5675e+6 3.1328e+6 4.6981e+6 6.2634e+6 7.829e+6

04

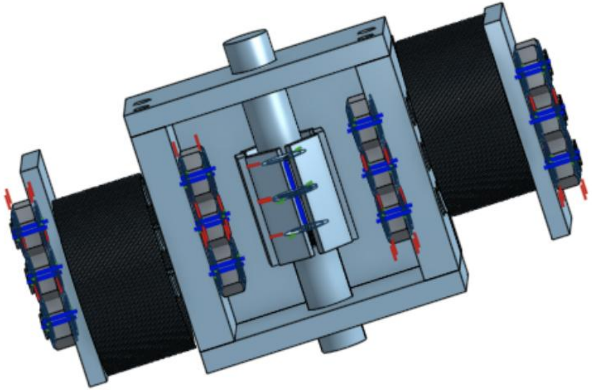
Electrical Design

AC GENERATOR

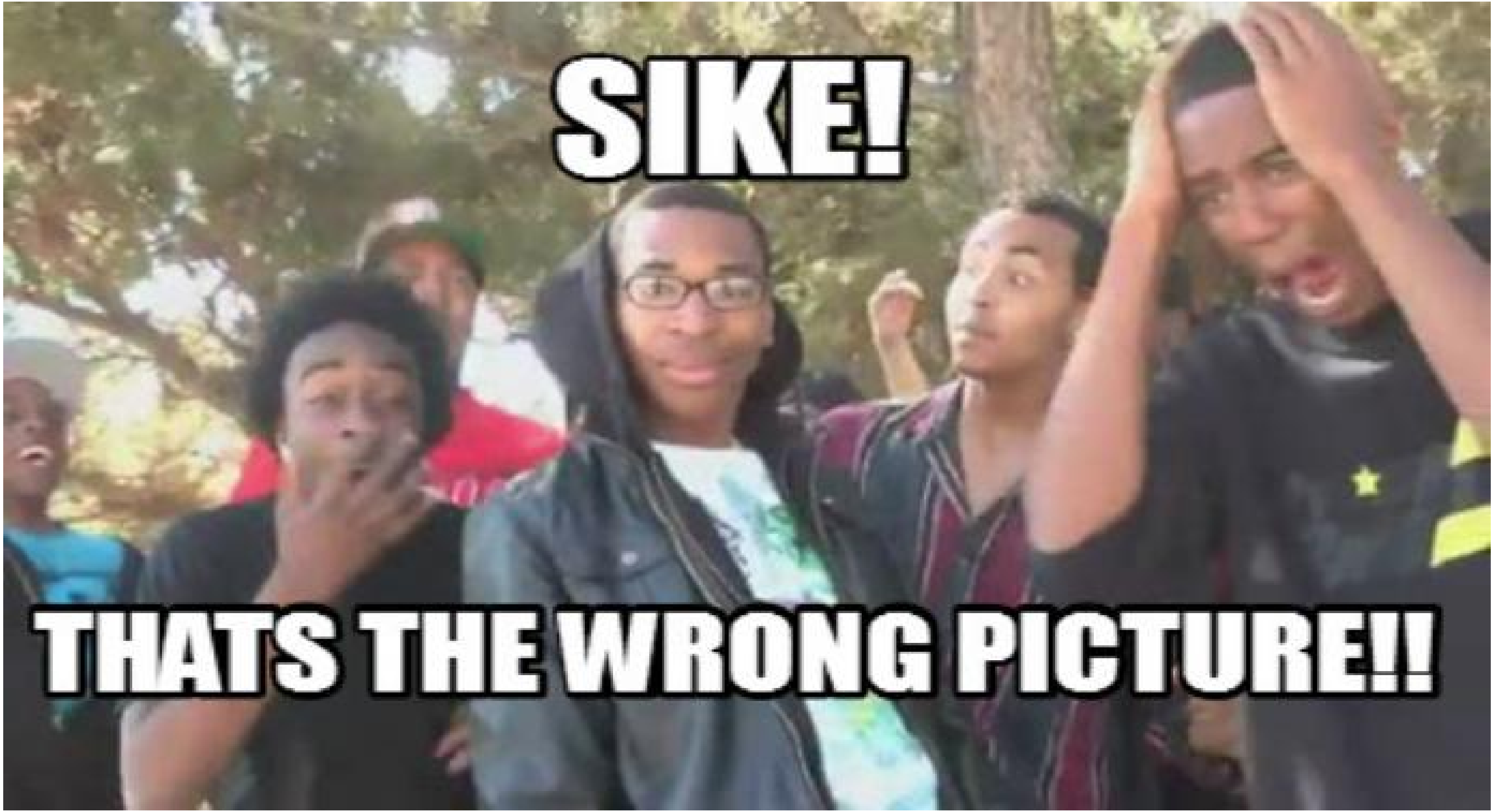
Premature Designs



v1

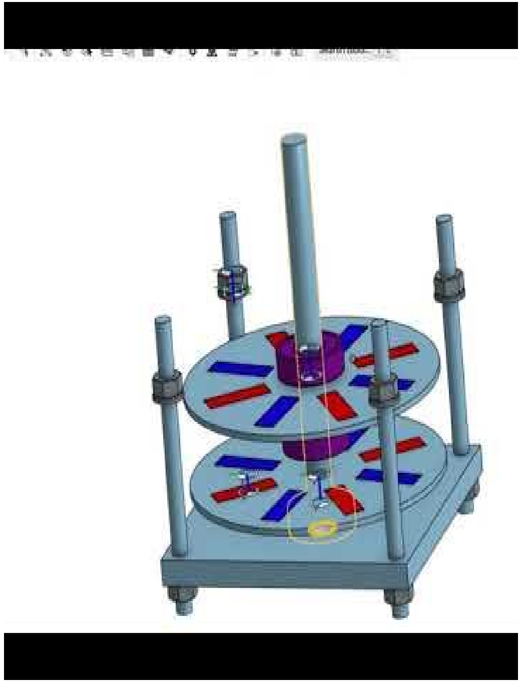


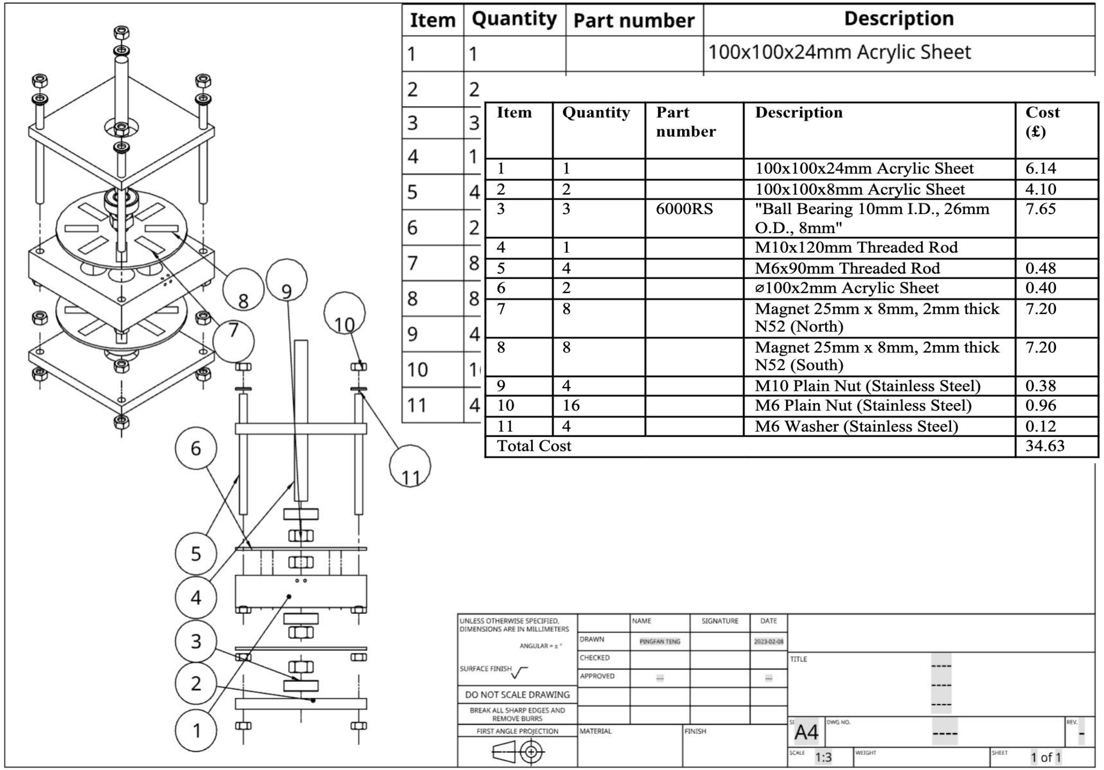
v2


SIKE!
THATS THE WRONG PICTURE!!

AC GENERATOR

Final Design





Item	Quantity	Part number	Description
1	1		100x100x24mm Acrylic Sheet
2	2		
3	3		
4	1		
5	4		
6	2		
7	8		
8	8		
9	4		
10	1		
11	4		

Item	Quantity	Part number	Description	Cost (£)
1	1		100x100x24mm Acrylic Sheet	6.14
2	2		100x100x8mm Acrylic Sheet	4.10
3	3	6000RS	"Ball Bearing 10mm I.D., 26mm O.D., 8mm"	7.65
4	1		M10x120mm Threaded Rod	
5	4		M6x90mm Threaded Rod	0.48
6	2		⌀100x2mm Acrylic Sheet	0.40
7	8		Magnet 25mm x 8mm, 2mm thick N52 (North)	7.20
8	8		Magnet 25mm x 8mm, 2mm thick N52 (South)	7.20
9	4		M10 Plain Nut (Stainless Steel)	0.38
10	16		M6 Plain Nut (Stainless Steel)	0.96
11	4		M6 Washer (Stainless Steel)	0.12
Total Cost				34.63

https://www.youtube.com/watch?v=YTzgTmnWmBQ

AC GENERATOR

Conditions and

Generator Input:

- 2900 RPM assuming 100% efficiency from the gears.

Generator Output:

- 10V and 200mA
- Some voltage used for rectification (1.4V)
- 50Hz signal

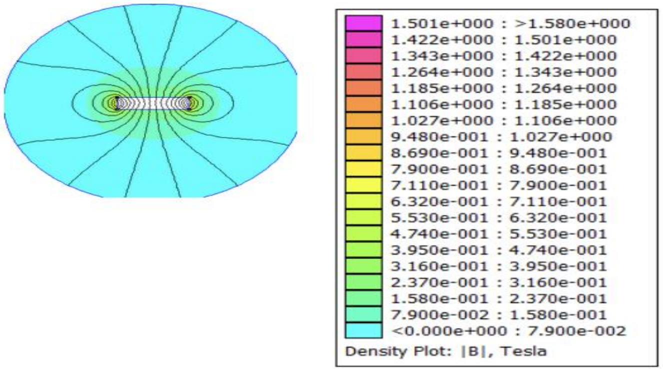
Single- Phase, 8 pole Stator Generator , Air gap < 2 mm
Estimated Efficiency 55%

Ns = 60 * f / Pole pairs

Centerline Gauss Calculator for a Rectangle Magnet

Magnet 1

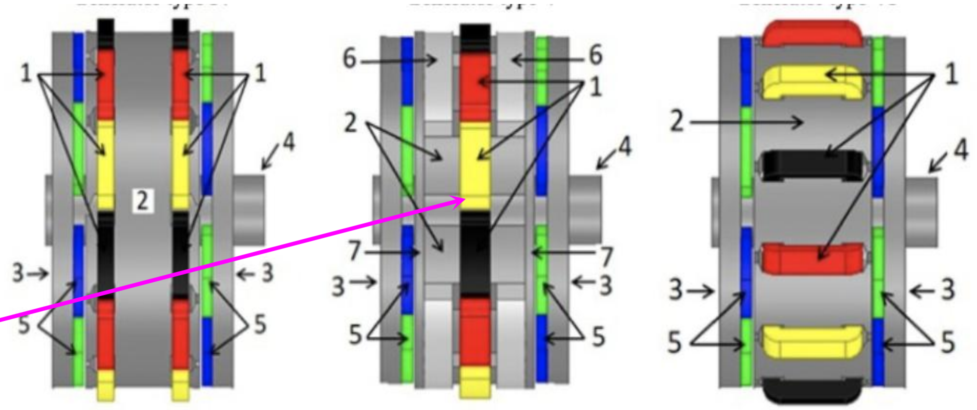
Units (inches or mm)	Inches
Width:	8
Length: (Magnet's length)	25
Thickness: (Magnet's thickness along magnetization)	2
Distance: (Distance from the magnets pole face (>0))	2
Material:	NdFeB
Grade:	NdFeB 52 MGOe
	Calculate
Bz: (Gauss)	1618.0



AC GENERATOR

Analysis

Generator	Permanent magnet	V_t ≥ 15 (V)	VR ≤ 10 (%)	THD ≤ 8 (%)	P_{out} ≥ 300 (W)	η →100 (%)
Type I	Circular	14.74	7.04	4.15	295.48	90.73
	Rectangular	16.93	6.13	2.01	340.96	91.56
	Trapezoidal	17.77	5.85	6.00	355.62	91.82
Type II	Circular	15.60	6.66	5.83	312.61	90.53
	Rectangular	18.29	5.68	8.69	367.51	91.46
	Trapezoidal	18.67	5.57	13.11	375.43	91.53
Type III	Circular	15.23	6.82	6.63	307.53	90.03
	Rectangular	18.77	5.86	9.13	360.63	90.86
	Trapezoidal	18.02	5.76	12.91	366.01	90.99
Type IV	Circular	15.16	6.85	6.37	306.10	89.96
	Rectangular	17.69	5.87	7.93	358.84	90.79
	Trapezoidal	17.98	5.78	12.60	364.65	90.91
Type V	Circular	15.70	6.61	6.11	314.85	90.16
	Rectangular	18.32	5.67	8.23	367.33	90.79
	Trapezoidal	18.72	5.55	12.75	375.43	90.88
Type VI	Circular	15.72	6.61	6.22	314.49	90.51
	Rectangular	18.42	5.64	8.50	369.50	91.36
	Trapezoidal	18.76	5.54	12.95	376.56	91.47
Type VII	Circular	15.71	6.61	6.49	314.86	90.42
	Rectangular	18.37	5.65	8.72	368.69	90.96
	Trapezoidal	18.74	5.54	13.05	376.28	91.07
Type VIII	Circular	11.76	8.83	13.71	234.94	65.89
	Rectangular	14.93	6.96	13.34	296.92	65.07
	Trapezoidal	15.09	6.88	14.03	300.22	65.20
Type IX	Circular	18.14	5.73	12.17	367.00	91.37
	Rectangular	20.45	5.08	7.14	410.18	91.93
	Trapezoidal	21.86	4.75	12.54	437.91	92.21



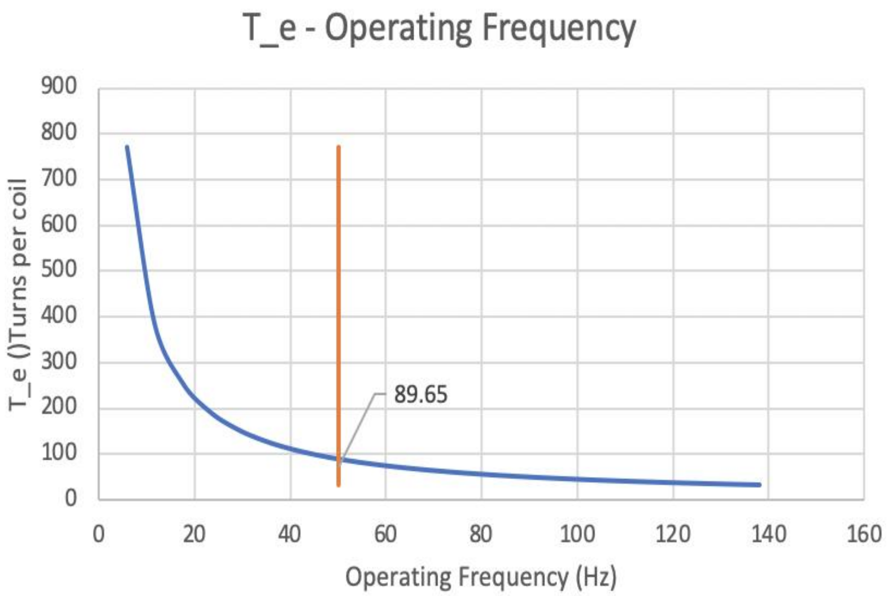
Generator type VII Generator type VIII Generator type IX

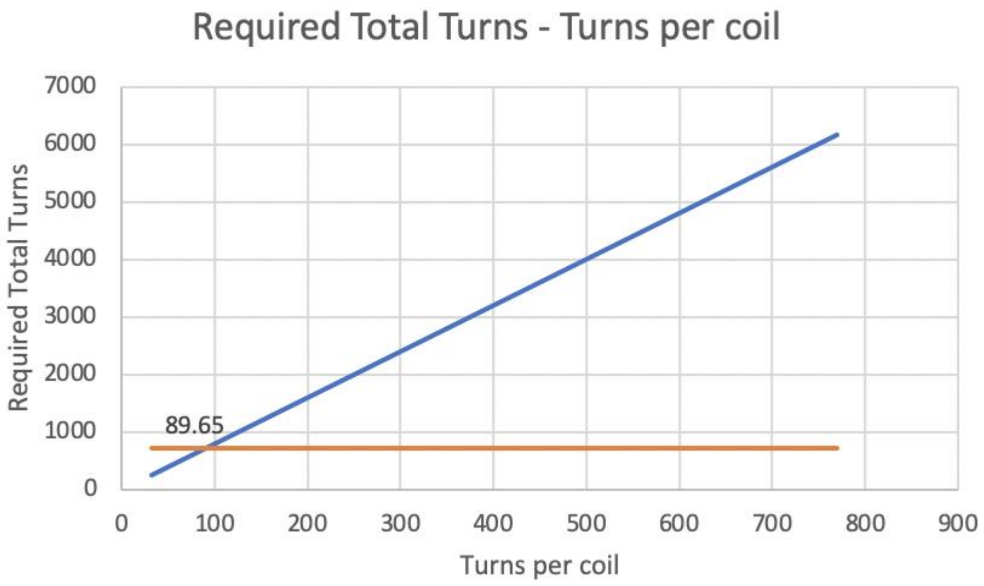
Remarks : 1- stator winding, 2- stator core, 3- rotor core, 4-shaft, 5- permanent magnet north and south, 6- plastic ABS, 7- stator slot opening closer.

AC GENERATOR

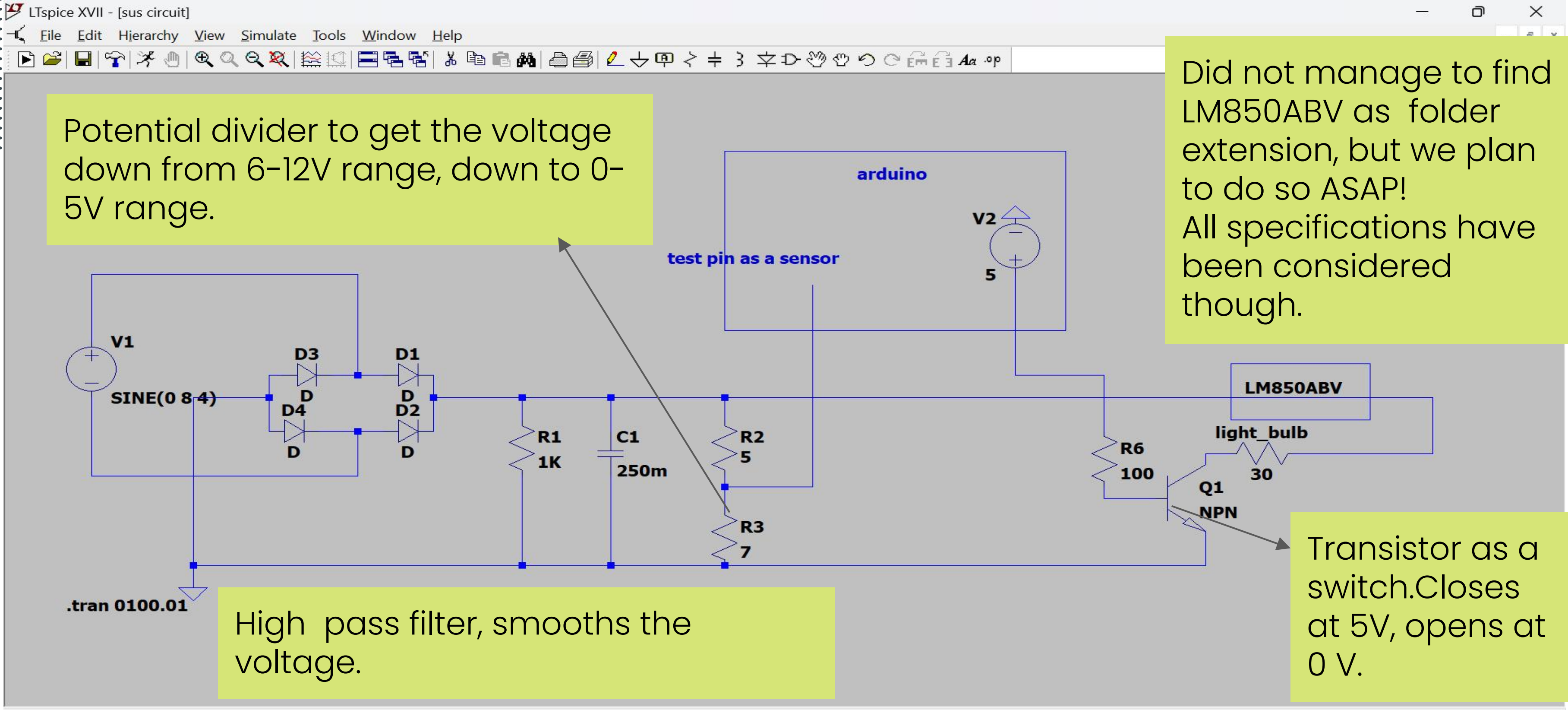
Further Analysis

Detailed Excel spreadsheet is available on Microsoft Teams - Channel: Electrical Parts - Files- Folder: Axial Flux Generator.





LT -SPICE CIRCUIT



Potential divider to get the voltage down from 6-12V range, down to 0-5V range.

Did not manage to find LM850ABV as folder extension, but we plan to do so ASAP! All specifications have been considered though.

High pass filter, smooths the voltage.

Transistor as a switch.Closes at 5V, opens at 0 V.

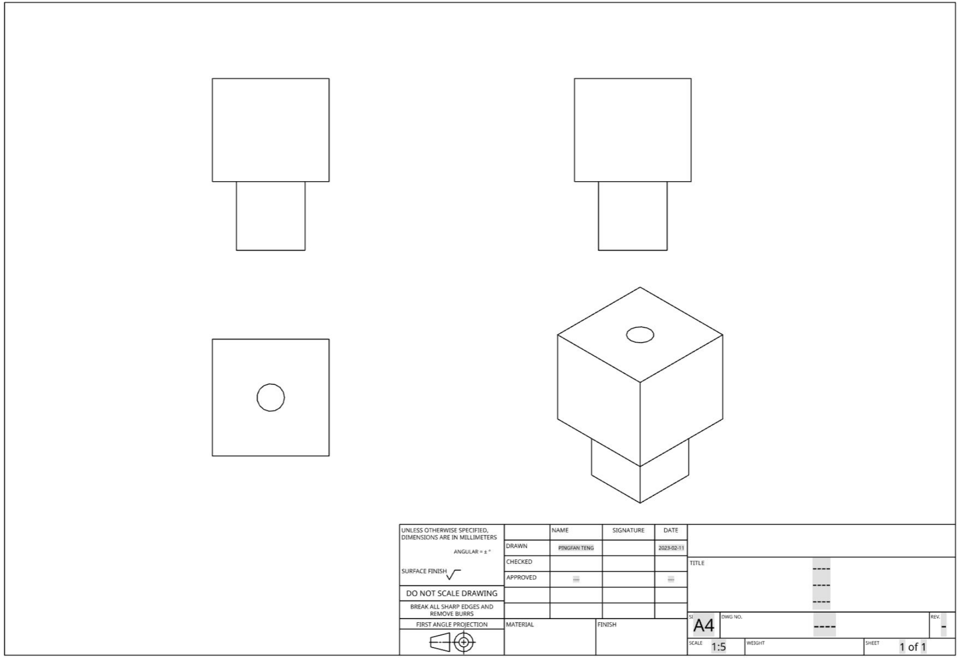
Arduino pseudocode

-Void set up (executed once)
- Start serial communication with a baud rate of 9600
- Set pin A3 as INPUT and pin 4 as OUTPUT

-Void loop (While loop of arduino)
- Read the voltage value from pin A2
- If the voltage value is between x and y ((inclusive):
  - For each index in the message array (the number of 0s or 1s in the morse code for the text "its windy"):
    - Print the voltage value to the serial monitor
    - Wait for 200 milliseconds
    - Read the voltage value from pin A2
    - If the voltage value is not between x and y:
      - Turn the output pin to LOW
      - Exit the loop
    - Else if the value at the current position of the message array is 1:
      - Turn the output pin to HIGH
    - Else:
      - Turn the output pin to LOW
- Else: (What to do if initial voltage is invalid)
  - Print the voltage value to the serial monitor
  - Turn the output pin to LOW
  - Wait for 200 milliseconds

Housing

Aluminium Box, with an insulated layer of plywood at the inner surface (Glued). But this will be adjusted based on the budget!



Bring the force we are prepared!

References

Generator Design:
https://www.sciencedirect.com/science/article/pii/S1110016821004300
Mechanical Engineer design MEME:

https://www.google.com/url?sa=i&url=https%3A%2F%2Fnewengineer.com%2Fadvice%2F10-best-memes-about-engineering-1259375&psig=AOvVaw2tbhqOtRJ8YZz1Yd5HoPHX&ust=1676211303284000&source=images&cd=vfe&ved=0CA4QjRxqFwoTCIjf9uvTjf0CFQAAAAAdAAAAABAo

Electrical design MEME:
https://www.google.com/url?sa=i&url=https%3A%2F%2Fm.facebook.com%2FElecEngMemes%2F&psig=AOvVaw0a7wj8hEO3DNc7OGpap9Wh&ust=1676210856270000&source=images&cd=vfe&ved=0CA8QjRxqFwoTCPD9ppfSjf0CFQAAAAAdAAAAABAh

SIKE THAT'S THE WRONG PICTURE MEME
https://www.google.com/url?sa=i&url=https%3A%2F%2Fwww.mememaker.net%2Fmeme%2Fsike-thats-the-wrong-picture%2F&psig=AOvVaw3D7crhjw3gw2UEoq1KU-Tl&ust=1676217104109000&source=images&cd=vfe&ved=0CA4QjRxqFwoTCNDA8rnpjf0CFQAAAAAdAAAAABAJ

References

I see wait Wuuuut? MEME
https://www.google.com/url?sa=i&url=https%3A%2F%2Fimgflip.com%2Fi%2F3fdeio&psig=AOvVaw3GtXoEx3oon3yOfoZRFq7M&ust=1676216559281000&source=images&cd=vfe&ved=0CA8QjRxqFwoTCMCSr7bnjf0CFQAAAAAdAAAAABAE

Gear Catalog:
https://khkgears2.net/catalog2/

Gear image
https://www.google.com/url?sa=i&url=https%3A%2F%2Funsplash.com%2Fs%2Fphotos%2Fgears&psig=AOvVaw0x4ZSp0D9pJfmOFOmnui0D&ust=1676381713224000&source=images&cd=vfe&ved=0CA4QjRxqFwoTCKjC8tXOkv0CFQAAAAAdAAAAABAI

https://www.adamsmagnetic.com/gauss-and-pull-calculators-magnets/.

https://www.sciencedirect.com/science/article/pii/S1110016821004300

https://www.youtube.com/watch?v=XU0uTMhFsis

https://forum.arduino.cc/t/resistor-with-6v-led/108371/4
https://www.sciencedirect.com/science/article/pii/S1110016821004300

Thank You!

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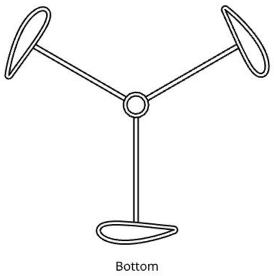
Examiner be like:

Wut!? 5V DC output? What about the 6V to light up the bulb?

1) Step-Up Module to adjust the voltage to 6V or
2) Let it be 5V and hope for the best!
   (Research shows light will just be dimmer)



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